

llmXive Automated Review of SynthDocBench: Controlled Benchmark for Long-Context Visual Document Understanding

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents a rigorous synthetic benchmark, but several specific quantitative claims in the abstract and main text do not align with the data presented in the results tables.

First, the abstract and Section 5 state that “five of six models show a negative Early→Late trend.” However, Table 5 (Positional Bias) lists eight models. Of these eight, only four (Gemini-3.1-Pro, GPT-4o, Claude-Sonnet-4.5, Qwen3.5-VL-122B) exhibit a negative Δ (Late - Early). Two models (GPT-5.4 and Qwen3-VL-235B) show positive trends, and two (InternVL3-78B, Qwen2.5-VL-7B) show near-zero or slightly positive trends. The claim “five of six” is factually unsupported by the provided table, which shows 4 of 8. This appears to be a copy-paste error from an earlier draft or a miscalculation.

Second, the model name “GPT-5.4” appears in Section 4 and Table 4, but the bibliography entry `openai2025gpt5` refers to “GPT-5” without a version suffix, and standard OpenAI naming conventions do not typically use “.4” for a major release in this context. Unless “GPT-5.4” is a specific internal version distinct from the public “GPT-5” cited, this is a citation drift or hallucinated version number that needs verification.

Finally, the abstract mentions “five of six models” again regarding the trend, reinforcing the inconsistency. The text must be corrected to reflect the actual count from Table 5 (4 of 8) or specify which subset of models was intended. These are not statistical errors but factual mismatches between the narrative and the reported data.

Action items.

- **[writing]** Abstract claims ‘five of six models show a negative Early→Late trend’ but Table 5 (positional bias) shows GPT-5.4 (+0.028) and Qwen3-VL-235B (+0.019) have positive trends. Only 4 of 8 models show negative trends. Correct the count to ‘four of eight’ or remove the specific ‘five of six’ claim.
- **[writing]** Section 4 lists ‘GPT-5.4’ as a baseline model, but the bibliography cites ‘openai2025gpt5’ (GPT-5) and the text elsewhere refers to ‘GPT-5’. The version number ‘.4’ appears unsupported by the citation or standard naming conventions. Verify the exact model version used and align the text and citation.
- **[writing]** Section 5 claims ‘five of six models’ show a negative trend, but the table includes 8 models. The abstract also says ‘five of six’. The denominator is inconsistent with the data presented in Table 5 (which has 8 rows). Clarify the subset of models being referenced or correct the count to match the full table.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure provides a clear high-level overview of the evaluation pipeline, but the visual depiction of single-page conversion contradicts the caption’s description of 5-page concatenated strips, and the specific judge model name is omitted from the diagram.

Figure 2

The figure effectively visualizes score distributions for most models, but the Qwen2.5-VL-7B entry is missing its box plot components, and the legend overlay obscures a critical x-axis value.

Figure 3

The figure attempts to show performance by evidence position but fails to map the legend to the data correctly. The legend lists 8 models while only 6 bars are plotted, and multiple models share identical colors, rendering the data uninterpretable.

Figure 4

Figure 4 clearly displays accuracy by question subset with model differentiation via color, but lacks error bars to indicate uncertainty and could improve legend clarity by explicitly linking colors to models.

Figure 5

The bar chart effectively displays the frequency of the top chart types, but it omits two categories mentioned in the caption (24 total types vs. 22 bars shown), creating a discrepancy between the text and the data presented.

Figure 6

The figure contains a title-axis contradiction and illegible axis labels that undermine data interpretation, while the caption’s reference to external figures is not visually supported in the image.

Figure 7

Figure 7 is a clear, well-structured flowchart that effectively visualizes the synthetic document generation pipeline described in the caption. All five stages are distinct, the flow direction is logical, and the internal components are legible and consistent with the textual description.

Figure 8

The figure provides a high-level overview of the QA generation pipeline but lacks clear visual indicators for data flow between the extraction and generation stages, and the feedback loop's mechanics are not explicitly defined.

Figure 9

Figure 9 presents document statistics but has inconsistent scale interpretation in the word count panel and ambiguous axis labels that reduce clarity.

Figure 10

Figure 10 contains multiple inconsistencies between the visual data and the caption claims regarding balance, mean values, and the number of chart types displayed.

Figure 11

The bar chart clearly displays accuracy values for two model variants across four subsets, with a legend and axis labels. However, the absence of error bars or confidence intervals limits the ability to evaluate the reliability of the observed performance differences.

Figure 12

The heatmap effectively visualizes performance across categories and models with a clear color scale and numerical values. However, the caption contains a broken reference to the appendix table, and the rotated x-axis labels are slightly difficult to read.

Action items.

- **[science]** Figure 1: The diagram depicts a single PDF page being converted to an image, whereas the caption states that pages are ‘grouped into concatenated 5-page strips’; the visual representation contradicts the described input format.
- **[science]** Figure 1: The caption specifies that the judge model is ‘GPT-5’, but this model name is not labeled in the diagram (only ‘Judge LLM’ is shown), creating a disconnect between the visual schematic and the specific methodology described.
- **[science]** Figure 2: The ‘Qwen2.5-VL-7B’ row displays only outlier markers (circles) and a minimum whisker, but lacks a visible box (interquartile range) and median line, making the distribution incomparable to other models.
- **[writing]** Figure 2: The legend ‘ACC threshold (6)’ is rendered as a semi-transparent white

box that obscures the x-axis tick mark and label for ‘6’, reducing readability.

- **[science]** Figure 3: The legend lists 8 models (including GPT-5.4 and Qwen3-VL-235B), but the chart only displays 6 bars per group. The legend entries for GPT-5.4 and Qwen3-VL-235B are not represented in the data visualization.
- **[science]** Figure 3: The legend contains duplicate colors for distinct models (e.g., two distinct blue entries for ‘Claude-Sonnet-4.5’ and ‘Qwen3.5-VL-122B’, and two distinct teal entries for ‘Gemini-3.1-Pro’ and ‘Qwen3-VL-235B’), making it impossible to distinguish which bar corresponds to which model.
- **[science]** Figure 4: The x-axis labels (‘Chart’, ‘Complex’, ‘Cross-Modal’) do not match the caption’s description of ‘question subset’ if the caption implies a different categorization; however, the labels are clear. The main issue is the lack of error bars or confidence intervals for the accuracy scores, which are single point estimates without indication of variance or statistical significance.
- **[writing]** Figure 4: The legend uses color swatches but does not explicitly state that each color corresponds to a different model; while the model names are listed, the mapping between color and model is only implied by the bar colors in the chart, which could be clarified with a direct legend entry per model.
- **[science]** Figure 5: The caption states the corpus covers 24 distinct types, but the chart displays 22 bars. The two missing types (likely Sankey and Radar, given the caption’s examples) are not accounted for in the visualization.
- **[science]** Figure 6: The chart title reads ‘2015-2022’ but the x-axis extends to 2023, creating a contradiction between the title and the data range shown.
- **[science]** Figure 6: The y-axis label ‘Formal Employment Rate (%)’ is illegible due to low resolution, making it impossible to verify the scale or units visually.
- **[writing]** Figure 6: The caption references ‘Figure 15’ and ‘Figure 18’ as examples, but the image shown is a composite of two unrelated charts without clear labeling of which is which.
- **[writing]** Figure 8: The caption describes the pipeline as parsing reports into ‘structured evidence channels’ and extracting key information, but the figure visually depicts these as two distinct, disconnected stages (Panel a and Panel b) without showing the data flow or connection between them.
- **[science]** Figure 8: The ‘Autocorrection’ and ‘Adversarial Verification’ loop is shown as a feedback mechanism, but the diagram lacks arrows indicating the direction of data flow or the specific inputs/outputs for these steps, making the process ambiguous.
- **[science]** Figure 9: The middle panel’s x-axis (Word count) extends to 20,000, yet the caption states the corpus spans 38–65 pages; a 20k-word document is inconsistent with this page range for standard reports, suggesting a potential labeling error or outlier skew not addressed.
- **[writing]** Figure 9: The middle panel’s x-axis label ‘Word count’ is missing units (e.g., ‘words’), while the other panels implicitly use counts; add explicit units for clarity.

- **[writing]** Figure 9: The right panel’s x-axis label ‘Charts’ is ambiguous—does it mean ‘number of charts per document’? Clarify in the axis label or caption.
- **[science]** Figure 10: Subplot (a) shows counts of 597, 597, and 594, which are not ‘balanced equally’ as claimed in the caption; the discrepancy should be explained or the claim corrected.
- **[science]** Figure 10: Subplot (c) histogram shows a mean of 51, but the caption states the mean is 50.8; the values should match for consistency.
- **[science]** Figure 10: Subplot (d) lists only 10 chart types, but the caption claims 24 distinct types are shown; the figure does not match the description.
- **[science]** Figure 11: The caption claims ‘Vision dominates on Chart’, but the chart shows the blue bar (GPT-4o vision, 0.46) is higher than the orange bar (OCR + GPT-4o, 0.30). However, the orange bar is labeled ‘0.30’ while the blue bar is labeled ‘0.46’, which contradicts the visual height where the blue bar is clearly taller. Wait, looking closer, the blue bar is indeed taller. The issue is the caption says ‘Vision dominates on Chart’ which matches the data ($0.46 > 0.30$). But the caption also says ‘OCR
- **[writing]** Figure 12: The caption states ‘Full numerical values are in Table (Appendix)’, but the specific table number or appendix section is missing, making the reference unresolvable.
- **[writing]** Figure 12: The x-axis labels (model names) are rotated at a steep angle, which reduces readability and makes it difficult to quickly identify the models.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper is generally well-structured regarding terminology, with a dedicated notation glossary in the appendix. However, there are specific instances where symbols are overloaded or defined in a way that creates friction for a reader not immediately consulting the appendix.

The most significant issue is the overloaded symbol τ . In Section 4 (Evaluation Setup), τ is explicitly defined as the accuracy threshold ($\tau=6$). However, in Appendix A (Table notation_appendix), τ is defined as the “Topic seed used to initialize document generation.” This collision is confusing because the symbol appears in the main text with a meaning that contradicts its formal definition in the appendix. A reader following the main text will assume τ is the threshold, but if they check the appendix for a “complete” list of symbols, they will find a conflicting definition. This requires a fix: either rename the topic seed variable in the appendix (e.g., to τ_{seed} or θ_{seed}) or explicitly distinguish the threshold symbol in the main text (e.g., τ_{acc}) and ensure the appendix reflects the primary usage.

Additionally, in Section 3, the introduction of V_k as the metadata object for the k -th visualization lacks immediate context regarding the index k . While the notation is standard, explicitly

stating that k ranges over the set of visualizations in the current document $\mathcal{D}$ would prevent any ambiguity about whether k is a global index or a local one.

Finally, the term “Early→Late trend” in Section 5 is used as a shorthand for a specific comparison of document thirds. While the table caption clarifies this, the prose itself should briefly unpack the term upon first use to ensure the reader understands the spatial partitioning being discussed without needing to cross-reference the table immediately.

These are minor fixes that significantly improve the self-containment of the paper for an adjacent-field reader.

Action items.

- **[writing]** Section 4, Eq. 1: The symbol au is used as a threshold ($au=6$) but defined in Appendix A as a ‘Topic seed’. This overloaded notation forces page-flipping. Define au as the accuracy threshold at first use in Section 4 and rename the topic seed symbol in the appendix (e.g., to au_{seed}).
- **[writing]** Section 3, ‘Grounded visualization synthesis’: The symbol V_k is introduced without explicitly stating that k indexes visualizations within document $\mathcal{D}$. Add a clause clarifying the index range (e.g., ‘where $k \in \{1, \dots, K\}$ indexes visualizations in $\mathcal{D}$ ’).
- **[writing]** Section 5, ‘Positional Bias’: The term ‘EarlyoLate trend’ is used without defining the bucket boundaries in the prose. Add a parenthetical definition at first use (e.g., ‘...negative EarlyoLate trend (comparing the first, middle, and last thirds of the document)...’).

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is generally sound, but there are three specific instances where the text makes quantitative claims that contradict the data presented in the paper’s own tables, or where the scope of a claim shifts without explanation.

First, in the **Abstract** and **Introduction**, the authors state that “five of six models” exhibit a negative Early→Late trend (positional bias). However, **Table 5** in Section 5 lists eight models. Examining the Δ (Late→Early) column:

- Gemini-3.1-Pro: -0.004 (Negative)
- GPT-5.4: +0.028 (Positive)
- GPT-4o: -0.046 (Negative)
- Claude-Sonnet-4.5: -0.117 (Negative)
- Qwen3.5-VL-122B: -0.160 (Negative)
- Qwen3-VL-235B: +0.019 (Positive)

- InternVL3-78B: +0.003 (Positive)
- Qwen2.5-VL-7B: +0.014 (Positive)

Only **three** of the eight models show a negative trend. Even if the authors intended to refer only to the “six” models mentioned in the abstract (likely excluding the two smallest open-weight models), the count of negative trends among the remaining six (Gemini, GPT-5.4, GPT-4o, Claude, Qwen3.5, Qwen3) is still only **three** (Gemini, GPT-4o, Claude). The claim “five of six” is mathematically inconsistent with the provided table. This needs a correction to the text to match the data (e.g., “three of eight models” or a re-evaluation of which models are being counted).

Second, there is a confusion regarding the role of **Claude-Sonnet-4.5**. In **Appendix B (Judge Validation)**, the text states: “Claude-Sonnet-4.5 is a systematically lenient judge... and was excluded on this basis.” This phrasing strongly implies that Claude was excluded from the study entirely. However, **Table 1** (Main Results) and **Section 4** explicitly list “Claude-Sonnet-4.5” as one of the eight candidate models evaluated. The exclusion applied only to its role as a *judge* (scoring model), not as a *candidate*. The text in the appendix should be rephrased to “excluded as a judge” to avoid the logical implication that the model was not part of the benchmark results.

Third, the **Abstract** and **Introduction** refer to “six models” when discussing the positional bias, while **Section 5** and **Table 5** present results for **eight models**. While this is a common editing artifact (updating the data but not the summary), it creates a logical gap where the premise (6 models) does not match the evidence (8 models). The summary sections should be updated to reflect the full set of evaluated models to ensure the conclusion follows from the presented evidence.

These are primarily writing-level fixes to align the narrative claims with the empirical data presented in the tables, rather than fundamental flaws in the experimental design or causal logic.

Action items.

- **[writing]** Section 5 claims ‘five of six models’ show a negative Early→Late trend, but Table 5 shows only 3 of 8 models (or 3 of 6 if excluding small models) have negative deltas. Correct the count in the text to match the table data.
- **[writing]** Appendix B states Claude-Sonnet-4.5 ‘was excluded,’ implying it wasn’t in the study, but Table 1 lists it as a candidate. Clarify that it was excluded only as a *judge*, not as a *candidate* model.
- **[writing]** The Abstract and Introduction refer to ‘six models’ for positional bias, while Section 5 and Table 5 present eight. Update the summary sections to reflect the full set of evaluated models for consistency.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper generally maintains a strong alignment between its synthetic, controlled experimental design and its claims about diagnostic capabilities. However, there are specific instances of overreach regarding the scope of the empirical findings, particularly in the quantification of “positional bias” and the implied efficacy of the benchmark.

1. Misrepresentation of Empirical Scope in Abstract and Conclusion The Abstract (Section 0) and Conclusion (Section 6) make specific quantitative claims about the prevalence of “positional sensitivity” that are not supported by the data in Table 5 (Section 5).

- **Claim:** The Abstract states, “five of six models show a negative Early→Late trend.”
- **Evidence:** Table 5 lists **eight** models, not six. Furthermore, calculating the Δ (Late - Early) column:
 - Gemini-3.1-Pro: -0.004 (Negative)
 - GPT-5.4: +0.028 (Positive)
 - GPT-4o: -0.046 (Negative)
 - Claude-Sonnet-4.5: -0.117 (Negative)
 - Qwen3.5-VL-122B: -0.160 (Negative)
 - Qwen3-VL-235B: +0.019 (Positive)
 - InternVL3-78B: +0.003 (Positive)
 - Qwen2.5-VL-7B: +0.014 (Positive)

Only **four** of the eight models show a negative trend (Gemini, GPT-4o, Claude, Qwen3.5). The claim “five of six” is factually incorrect regarding both the count of models and the direction of the trend for the majority. The Conclusion repeats a similar error (“four of six models”), again using the wrong denominator.

- **Fix:** Correct the counts to reflect the actual eight models evaluated and the specific number showing the trend (e.g., “four of eight models”). Ensure the text does not generalize a trend observed in a subset to the entire evaluated population.

2. Implied Causality in Conclusion The Conclusion states the benchmark provides signals “needed to identify, understand, and ultimately improve model capabilities.”

- **Gap:** The paper demonstrates that the benchmark *identifies* failure modes and correlates with other benchmarks. It does not provide evidence that using this benchmark *causes* or *leads to* the improvement of model capabilities. This is a common rhetorical leap in benchmark papers, but it overreaches the demonstrated evidence.
- **Fix:** Narrow the claim to the diagnostic utility. Change “ultimately improve” to “guide the development of” or “inform strategies to improve.”

3. Consistency of Model Counts Throughout the text, the authors inconsistently refer to the number of models evaluated (sometimes “six,” sometimes “seven,” sometimes “eight”). While the table clearly lists eight, the prose should consistently reflect the full scope of the experiment to avoid confusion about the generalizability of the findings. The “six” count appears to be a remnant of an earlier draft or a subset analysis that was not updated in the final text.

These issues are primarily rhetorical and can be resolved by tightening the language to match the precise data presented in the tables, without requiring new experiments.

Action items.

- **[writing]** Abstract claims ‘five of six models show a negative Early→Late trend,’ but Table 5 (Section 5) shows only 3 of 8 models (GPT-4o, Claude, Qwen3.5) have a negative Δ (Late-Early). GPT-5.4, Qwen3-VL, InternVL3, and Qwen2.5-VL show positive or near-zero trends. The claim overstates the prevalence of the trend and miscounts the models. Revise to ‘three of eight models’ or specify the subset.
- **[writing]** Abstract states ‘five of six models’ for positional sensitivity, but the study evaluates eight models (Table 4). The conclusion (Section 6) repeats ‘four of six models’ for the same metric. The denominator ‘six’ is inconsistent with the actual experimental scope (eight models) and the numerator counts are factually unsupported by the provided data. Align the text with the actual model count and observed trends in Table 5.
- **[writing]** Conclusion states the benchmark serves to ‘ultimately improve model capabilities,’ implying a direct causal link between the diagnostic signal and future performance gains. The paper only demonstrates diagnostic signal (correlation with MMLongBench-Doc, identification of failure modes). It does not demonstrate that using this benchmark leads to improved models. Rephrase to ‘provides signals to guide future model development’ to avoid implying unproven efficacy.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a low-risk research artifact: a fully synthetic benchmark for long-context visual document understanding. The authors explicitly state in the **Ethics Statement** (main.tex, lines 138–168) that the dataset is constructed entirely from programmatically generated synthetic documents, involves no human subjects, and contains no personally identifying information (PII) or real-world sensitive data. The content is grounded in broad, public topic seeds and reviewed to exclude hate speech or harmful material.

The methodology avoids the primary safety pitfalls associated with data collection: there is no scraping of private user data, no use of human-subject logs requiring IRB approval, and no release of re-identifiable information. The “dual-use” risk is minimal; while the benchmark

evaluates model capabilities in reading charts and documents, it does not provide a mechanism for generating disinformation, bypassing safety filters, or executing cyberattacks. The synthetic nature of the data actually mitigates the risk of the benchmark being used to train models on harmful real-world content.

The paper includes a dedicated **Reproducibility Statement** and **Ethics Statement** that address environmental impact, benchmark integrity (Goodhart’s Law), and intended use limitations. No specific, foreseeable, non-trivial risks of harm were identified that are unacknowledged or unmitigated in the text. The work is safe to publish.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a well-constructed synthetic benchmark with a clear factorial design, but the evidentiary support for two central claims requires strengthening to rule out plausible alternative explanations.

First, the headline result—that Gemini-3.1-Pro significantly outperforms other models (Table 1)—is potentially confounded by the benchmark’s generation pipeline. The authors acknowledge in Section 5 that the use of D3.js/HTML rendering might favor models with high exposure to web-rendered content (“rendering-familiarity”). However, the paper offers no empirical evidence to rule this out. The 13.9 percentage point gap could be entirely driven by the model’s familiarity with the specific visual style of D3.js charts rather than a genuine advantage in long-context reasoning. To support the claim that the benchmark measures reasoning capabilities, the authors must demonstrate that the performance gap persists when the visual style is altered. An ablation using a distinct rendering backend (e.g., LaTeX-generated PDFs or non-D3 rasterization) for a held-out subset of questions is necessary to isolate the reasoning effect from the rendering artifact.

Second, the claim of a systematic “positional bias” (Section 5, Table 3) relies on a single evaluation run per model. While the paper reports bootstrap confidence intervals for overall accuracy, it does not report variance for the positional bias metric itself (the drop in accuracy for the middle third). With only 200 documents, a 5–18 percentage point drop could plausibly arise from the specific distribution of question types or chart placements in that single sample, rather than a robust model failure mode. The evidence design fails to rule out sampling noise. The authors should report the positional bias metric across multiple random seeds or document splits, providing standard deviations or confidence intervals for the bias magnitude to confirm the effect is stable.

Finally, the comparison between the “OCR + Text-Only” baseline and the vision models (Section 4) suffers from a potential asymmetry in information access. The OCR baseline uses PyMuPDF to extract text, but it is unclear if this extraction includes structured table data or

layout metadata that the vision models are explicitly denied (as they receive only rasterized images). If the OCR baseline inadvertently accesses structured data that the vision models cannot, the conclusion that “visual perception is the primary bottleneck” is confounded. The baseline must be re-run with strict constraints ensuring it only has access to raw, unstructured text, matching the information constraints of the vision-only protocol.

Action items.

- **[writing]** The paper presents a well-constructed synthetic benchmark with a clear factorial design, but the evidentiary support for two central claims requires strengthening to rule out plausible alternative explanations. First, the headline result—that Gemini-3.1-Pro significantly outperforms other models (Table 1)—is potentially confounded by the benchmark’s generation pipeline. The authors acknowledge in Section 5 that the use of D3.js/HTML rendering might favor models with high exposure to web-rendered

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical treatment in this paper is largely sound, utilizing bootstrapping to estimate confidence intervals for a fixed benchmark size. However, there are three specific areas where the reporting of uncertainty and the interpretation of statistical patterns require refinement to meet rigorous standards.

First, the main results tables (Table 1 and Table 2) present accuracy and mean scores to three decimal places (e.g., 0.725, 7.19) without accompanying measures of uncertainty such as standard deviation, standard error, or confidence intervals within the table cells. While Appendix B provides bootstrap CI half-widths, the primary presentation of results suggests a level of precision that a single point estimate cannot support. To avoid false precision, the authors should either report the standard error of the mean derived from the bootstrap distribution alongside the point estimate in the main tables or explicitly state that these values are single-point estimates without calculated variance.

Second, the analysis of positional bias in Section 5 makes definitive claims such as “the middle third is hardest for 5 of 8 models” based on the descriptive data in Table 3. The authors count how many models exhibit a lower score in the middle bucket but do not perform a formal statistical test (such as a Friedman test or repeated-measures ANOVA) to determine if the variation across positions is statistically significant or distinguishable from random noise. The text should be rephrased to reflect that these are observed descriptive trends rather than statistically validated effects, or a formal non-parametric test should be added to support the claim.

Third, the statement that “all pairwise ACC gaps between adjacent-ranked models exceed their combined CI half-widths” is used to justify the reliability of rankings. While non-overlapping

95% CIs are a strong indicator of difference, they are not strictly equivalent to a hypothesis test at $\alpha = 0.05$ for the difference between two means (which requires the CI of the *difference*). Given the large gaps reported, the conclusion is likely robust, but the phrasing implies a formal test was conducted. The authors should either perform a bootstrap test on the difference of means or soften the language to indicate that the gaps exceed the sum of half-widths, suggesting distinct performance levels.

These issues are primarily related to reporting precision and the framing of descriptive statistics as inferential results, rather than fundamental flaws in the experimental design.

Action items.

- **[writing]** Tables 1 and 2 report ACC to 3 decimals without uncertainty metrics. Add mean $\pm$ SD/SE or explicitly state these are single-point estimates to avoid false precision.
- **[writing]** Section 5 claims ‘middle third is hardest’ based on descriptive counts in Table 3. Rephrase as observed trends or run a repeated-measures test (e.g., Friedman) to support statistical significance.
- **[writing]** The claim that ‘gaps exceed combined CI half-widths’ implies significance. Clarify that non-overlapping CIs are a conservative proxy, or run a bootstrap test on the difference of means.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured, but the abstract contains a critical editing error where two distinct drafts of the same text are pasted sequentially without merging. The first paragraph begins “Multimodal large language models...” and the second “Vision language models...”; both attempt to summarize the work but with inconsistent phrasing and slightly different result claims (e.g., the first mentions “six layout archetypes” while the second repeats it but adds different specific failure mode details). This forces the reader to stop and reconcile the redundancy. The first paragraph should be deleted entirely, retaining the more detailed second version.

In the Introduction, the first contribution item is a single, massive sentence (over 100 words) that attempts to define three distinct benchmark subsets (chart, cross-modal, complex) and their specific characteristics in one breath. This creates a garden-path sentence where the reader must hold multiple clauses in memory before reaching the end. Breaking this into three distinct sentences or a bulleted list would significantly improve readability.

In Section 5, the transition between the general claim about positional bias (“middle third is hardest for 5 of 8 models”) and the specific example of Qwen3.5-VL-122B is abrupt. A simple bridging phrase like “For instance,” would clarify that the specific model is an illustration of the general trend.

Finally, in Appendix A, the phrase “four candidate–judge pairs” is ambiguous without immediate context, as the text only explicitly lists two candidates and two alternative judges. Explicitly stating “two candidates evaluated under two alternative judges” would remove the need for the reader to perform mental arithmetic to verify the count. These are minor structural and phrasing issues that, once fixed, will allow the reader to move through the text without friction.

Action items.

- **[writing]** The paper is generally well-structured, but the abstract contains a critical editing error where two distinct drafts of the same text are pasted sequentially without merging. The first paragraph begins “Multimodal large language models...” and the second “Vision language models...”; both attempt to summarize the work but with inconsistent phrasing and slightly different result claims (e.g., the first mentions “six layout archetypes” while the second repeats it but adds different specific failure